Week 5 - Cloud and API deployment

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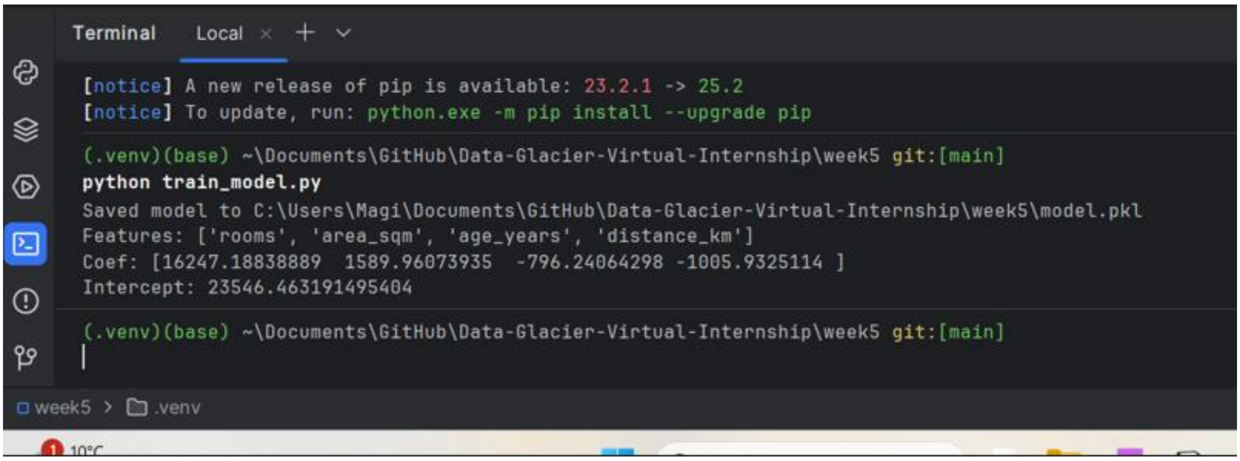
Introduction

The objective of this project was to train a simple House Price Prediction model using toy data (Week 4 dataset) and deploy it on the cloud. The deployment was implemented using AWS (EC2, ECR, IAM) and automated through GitHub Actions CI/CD.

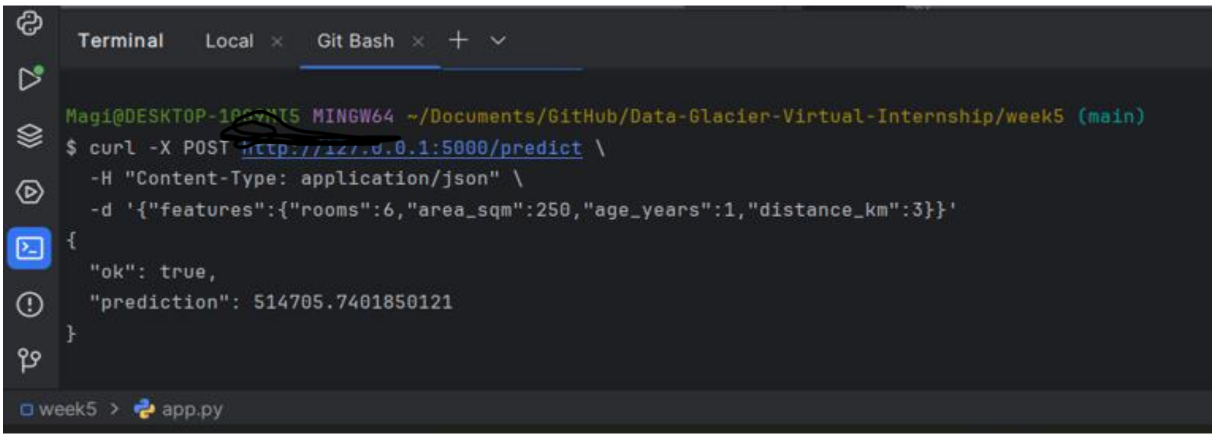
Step 1: Model Training

The model was trained on toy data and saved as model.pkl.



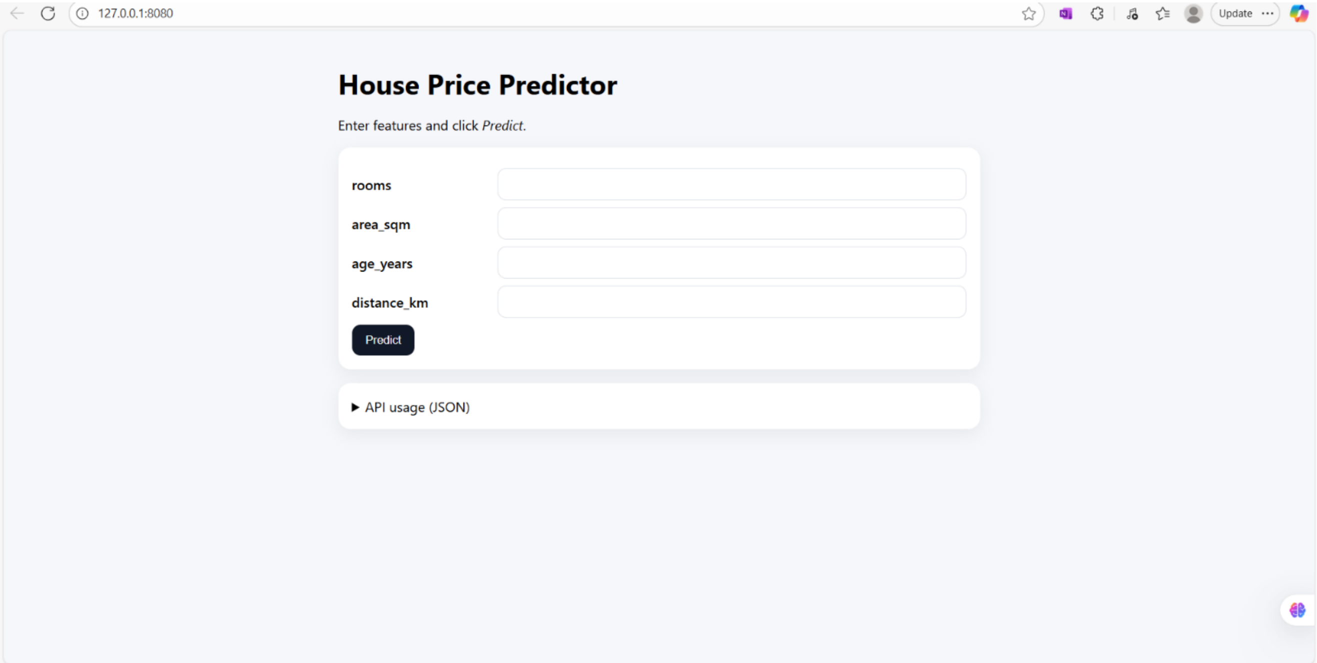
Step 2: Local API Test (JSON)

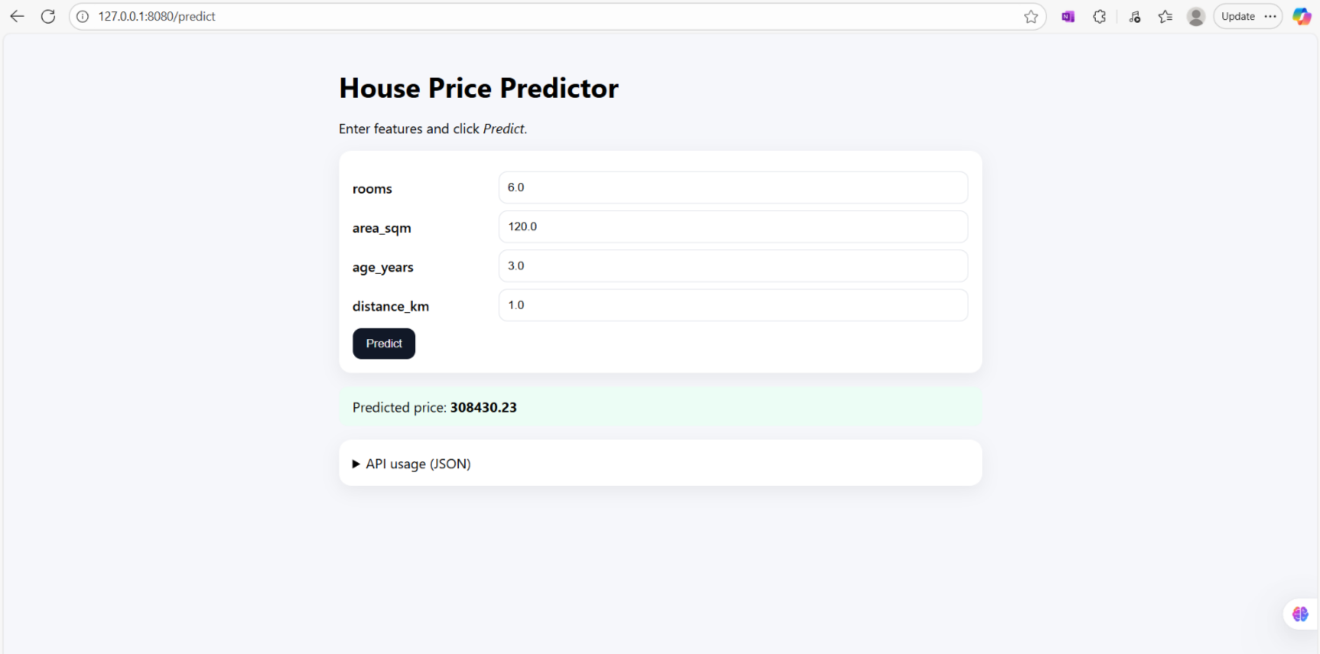
The API endpoint was tested locally using a curl command to verify predictions via JSON requests.



Step 3: Local Web Application

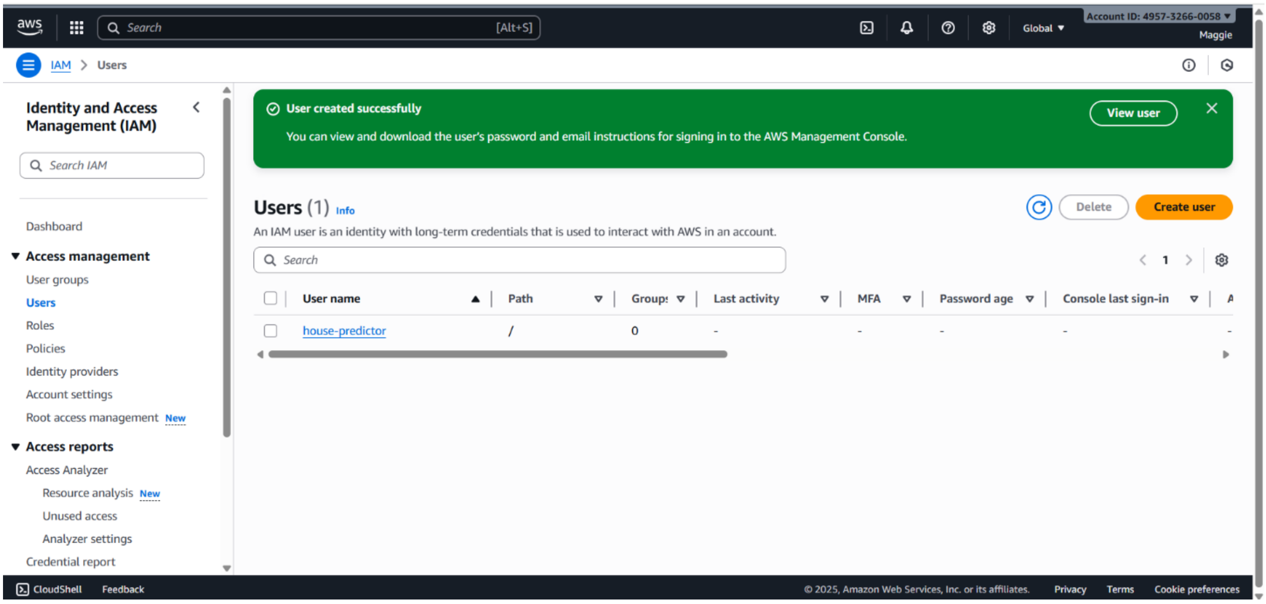
The Flask-based web application was run locally to test predictions from the user interface.





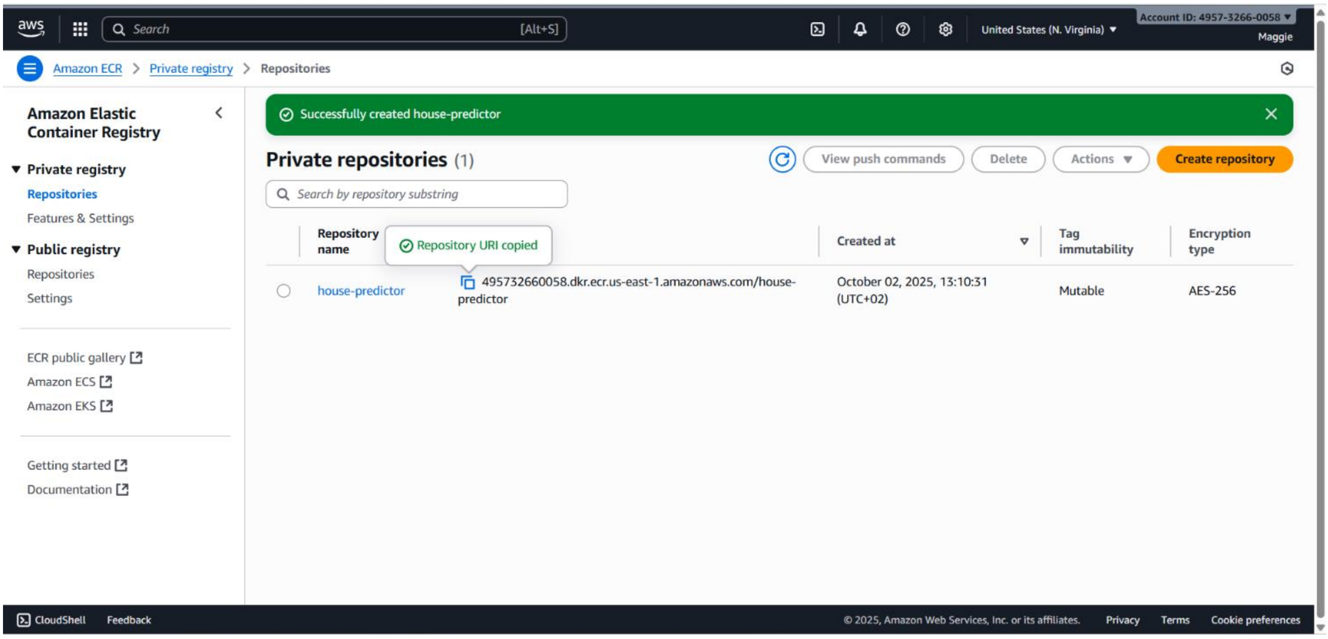
Step 4: AWS IAM Configuration

An IAM user was created with permissions to push Docker images to ECR and enable deployment.



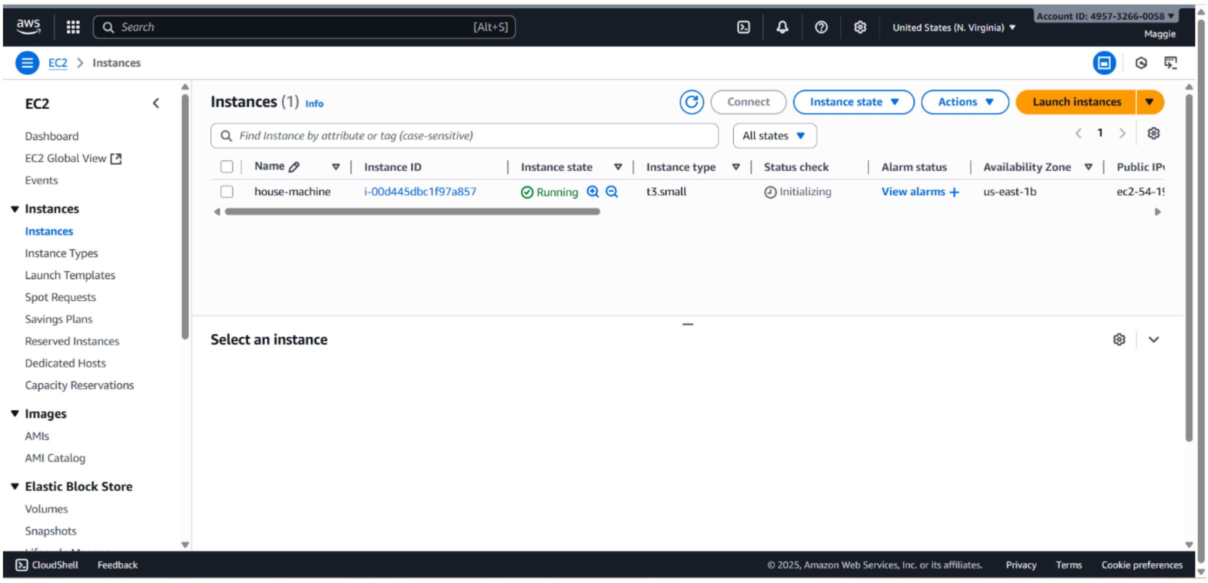
Step 5: AWS Elastic Container Registry

The Docker image for the application was pushed and stored in AWS ECR.



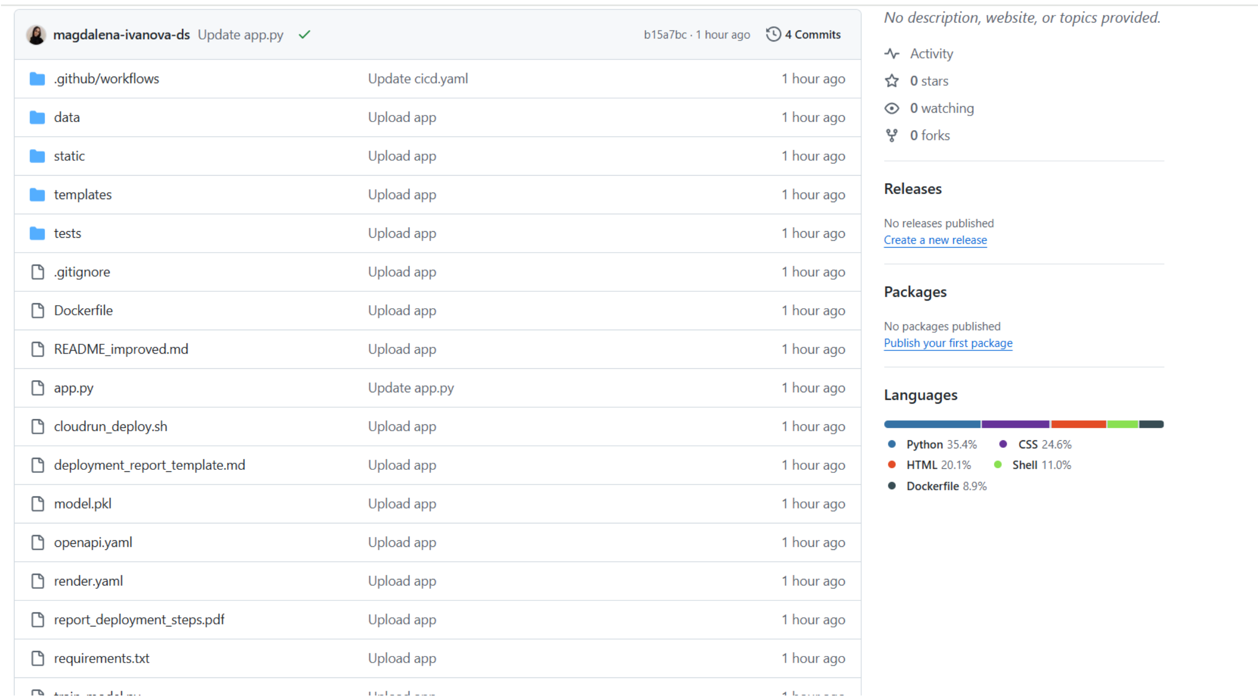
Step 6: AWS EC2 Instance

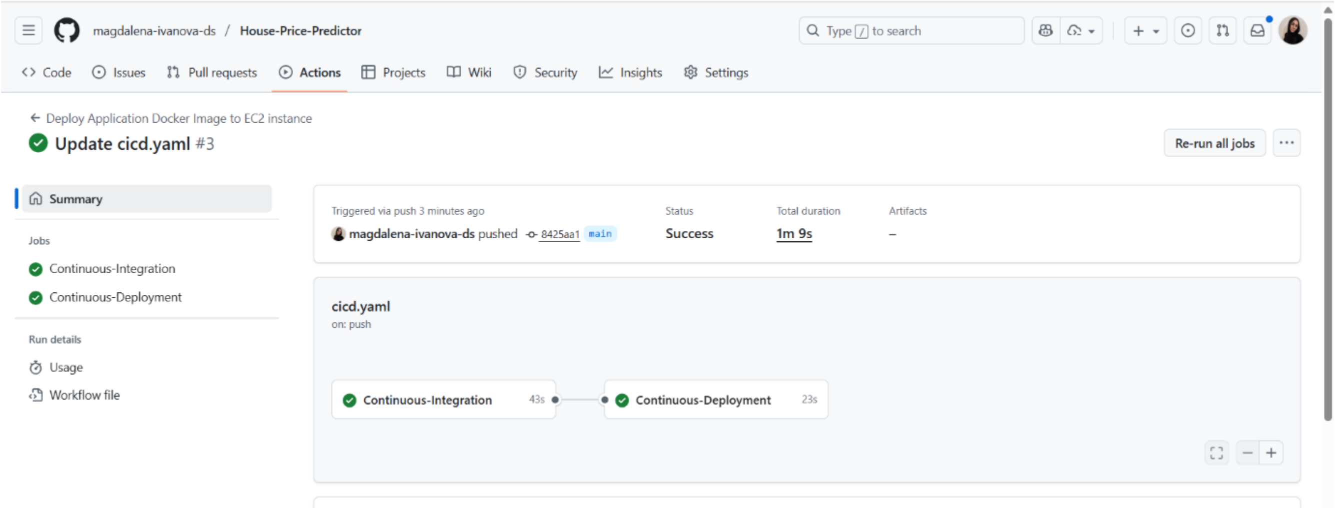
An EC2 instance was created to host the deployed Docker container.



Step 7: Continuous Integration / Deployment (GitHub Actions)

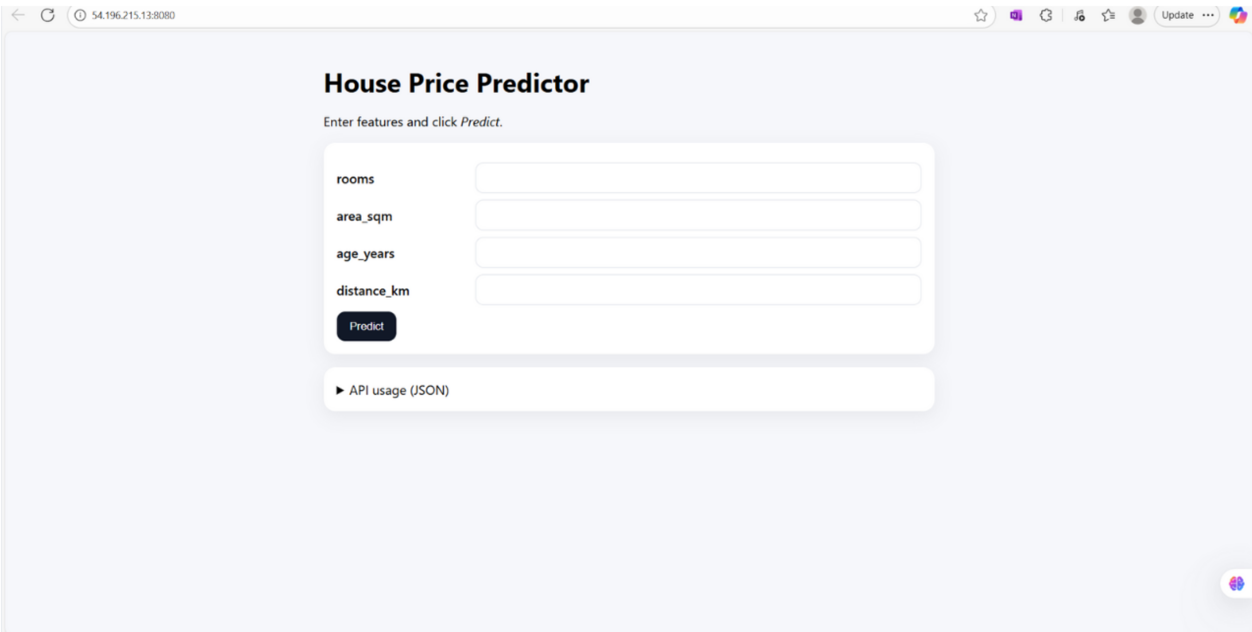
A CI/CD pipeline was configured with GitHub Actions to automate testing and deployment. (.github/workflows)





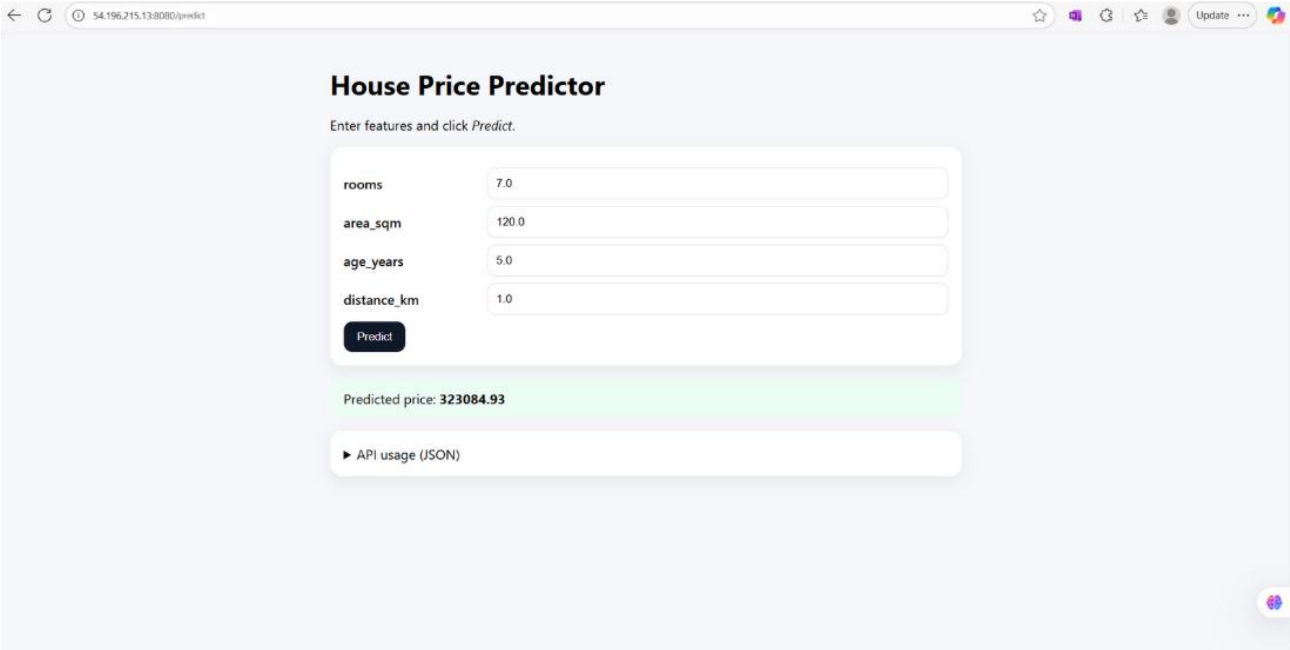
Step 8: Deployment on AWS EC2

The model and web application were deployed on AWS EC2 using Docker.



Step 9: Deployed Web Prediction

The deployed application was accessed on the web, and predictions were successfully generated.



Tools Used

- **Python 3.12** – Model training and API development
- **Flask** – Web framework for API and web app
- **Docker** – Containerization of the application
- **AWS EC2** – Cloud compute instance for deployment
- **AWS ECR** – Container registry for storing Docker images
- **AWS IAM** – Secure access management
- **GitHub Actions** – CI/CD automation

Conclusion

The model was successfully trained, tested locally, containerized, and deployed on AWS EC2 using Docker. GitHub Actions was integrated to enable continuous integration and deployment.